AI Model Card

AutoContour

Company: RadFormation | Generated: 10/23/2026

General Information

Product Name:	AutoContour
Company:	RadFormation
Version:	2.8
Category:	Auto-Contouring
Release Date:	3/19/2026
Last Updated:	9/26/2026
Description:	AutoContour is intended to assist radiation treatment planners in contouring and reviewing structures within medical images in preparation for radiation therapy treatment planning. (Source: FDA 510(k) K262452 Summary, accessed 2026-09-22)

Evidence Classification

Rigor E2 — External, independent or multi-center validation **Efficacy L3 — Decision/workflow efficacy**

Readiness R3 — Enhanced implementation readiness

+ Vendor-independent **- Multi-centre** **- Multi-national** **- Prospective** **+ External validation**

8 scored publication(s) · 8 listed · score origin: papers
Rigor: Independent multi-institutional evaluation (Goddard 2024, 5 facilities) plus Doolan 2023 and Fan 2025 support E2.
Efficacy: I2: Peer-reviewed studies support contour quality, physician acceptability, and workflow time savings, but also show anatomy/guideline-dependent failure modes requiring expert review and local QA. No prospective evidence of improved tumor control, toxicity, or survival was identified.
Readiness: Derived from E2 + CE MDR Class IIa + FDA 510(k): moderate implementation effort — local validation, interface testing, structure nomenclature mapping, and ongoing contour QA required before clinical use.

Key Features

- 520+ AI-trained models across CT, MR & CBCT (vendor-reported, v2.8; 523 distinct models on the product page: 435 CT, 81 MR, 7 CBCT)
- CT, MR & CBCT support; new MR abdomen models in v2.8
- v2.8 additions: mediastinal lymph node models, EPTN brain MR structures (e.g. caudate nucleus, fornix, pineal gland)
- Aligned with RTOG/NRG/RADCOMP/ESTRO/DAHANCA/EORTC/EPTN guidelines
- TG-263 nomenclature
- Eclipse integration & DICOM vendor-neutral

For information only. Not a clinical validation or endorsement; verify with the manufacturer and local commissioning.

- Planning Structures with Clean and Smooth post-processing (v2.8)
- Zero-Click queue dashboard with prioritisation and re-run (v2.8)
- Cloud or on-premises processing (parallel CPU, optional GPU)
- Deformable registration & re-irradiation workflows
- Unified AutoContour + Limbus Contour platform

Technical Specifications

Input Formats:	DICOM
Output Formats:	DICOM-RTSTRUCT
Processing Time:	Review-ready contours in seconds-to-minutes
Integration Options:	Eclipse (ESAPI); DICOM-compatible TPS import/export; Local Windows client/agent; Cloud-based automatic contouring service
Deployment Options:	Cloud-based automatic contouring service with local Windows client/agent
Target Population:	Adult patients
Modality Support:	CT; MRI; CBCT
Target Anatomy:	Brain; Head & Neck; Thorax; Abdomen; Pelvis; Breast
Disease Targeted:	Multiple Cancer Types

Supported Structures (607)

Head & Neck — 84

A_Carotid_L	A_Carotid_R	Body+Mask
Bone_Hyoid	Bone_Mandible	Bone_Pterygoid_L
Bone_Pterygoid_R	Bone_Skull	Bone_Teeth
Brain	Brainstem	BrachialPlex_L
BrachialPlex_R	BrachialPlexs	BuccalMucosa
Cavity_Oral	Cavity_Oral_Ext	Cerebellum
Clavicle_L	Clavicle_R	Cochlea_L
Cochlea_R	Cornea_2d_L	Cornea_2d_R
Cornea_L	Cornea_R	CorpusCallosum
Cribriform	Dental_Artifact	Ear_Internal_L
Ear_Internal_R	Esophagus	Eye_L
Eye_R	Eyes	Falx
Glnd_Lacrimal_L	Glnd_Lacrimal_R	Glnd_Submand_L
Glnd_Submand_R	Glnd_Thyroid	Hippocampus_L
Hippocampus_R	InternalAuditoryCanal_L	InternalAuditoryCanal_R
Larynx	Larynx_Glottic	Larynx_NRG
Larynx_SG	Lens_L	Lens_R
Lips	Lobe_Temporal_L	Lobe_Temporal_R
Macula_L	Macula_R	Medulla
Midbrain	Musc_Constrict_2d	Musc_Constrict
Musc_Sclmast_L	Musc_Sclmast_R	OpticChiasm

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Head & Neck — 84 (cont.)

OpticNrv_L	OpticNrv_R	Optics
Parotid_Glnd_L	Parotid_Glnd_R	Parotid_L
Parotid_R	Pharynx	Pituitary
Pituitary_3d	Pons	Retina_2d_L
Retina_2d_R	Retina_L	Retina_R
Sinuses	Tentorium	Trachea
V_Jugular_L	V_Jugular_R	Ventricle_Brain

Head & Neck Nodes — 55

LN_Neck_IA_2d	LN_Neck_IA_VI_2d	LN_Neck_II-IV_2d_L
LN_Neck_II-IV_2d_R	LN_Neck_CerVII_2d_L	LN_Neck_CerVII_2d_R
LN_Neck_2d_L	LN_Neck_2d_R	LN_Neck_VI_2d
LN_Neck_IA	LN_Neck_IB_2d_L	LN_Neck_IB_2d_R
LN_Neck_IB_L	LN_Neck_IB_R	LN_Neck_IB-V_L
LN_Neck_IB-V_R	LN_Neck_II_2d_L	LN_Neck_II_2d_R
LN_Neck_II_L	LN_Neck_II_R	LN_Neck_II-V_L
LN_Neck_II-V_R	LN_Neck_III_2d_L	LN_Neck_III_2d_R
LN_Neck_III_L	LN_Neck_III_R	LN_Neck_IV_2d_L
LN_Neck_IV_2d_R	LN_Neck_IV_L	LN_Neck_IV_R
LN_Neck_IVA_2d_L	LN_Neck_IVA_2d_R	LN_Neck_IVB_2d_L
LN_Neck_IVB_2d_R	LN_Neck_II-IV_L	LN_Neck_II-IV_R
LN_Neck_V_2d_L	LN_Neck_V_2d_R	LN_Neck_VC_2d_L
LN_Neck_VC_2d_R	LN_Neck_V_L	LN_Neck_V_R
LN_Neck_VIA	LN_Neck_VIIA_2d_L	LN_Neck_VIIA_2d_R
LN_Neck_VIIA_L	LN_Neck_VIIA_R	LN_Neck_VII_2d_L
LN_Neck_VII_2d_R	LN_Neck_VIIB_2d_L	LN_Neck_VIIB_2d_R
LN_Neck_VIIB_L	LN_Neck_VIIB_R	LN_Post_Neck_L
LN_Post_Neck_R		

Breast Nodes — 58

LN_Ax_L1_ESTRO_2d_L	LN_Ax_L1_ESTRO_2d_R	LN_Ax_L2_ESTRO_2d_L
LN_Ax_L2_ESTRO_2d_R	LN_Ax_L2_InPec_ESTRO_2d_L	LN_Ax_L2_InPec_ESTRO_2d_R
LN_Ax_L3_ESTRO_2d_L	LN_Ax_L3_ESTRO_2d_R	LN_IMN_ESTRO_2d_L
LN_IMN_ESTRO_2d_R	LN_IMN_Expand_ESTRO_2d_L	LN_IMN_Expand_ESTRO_2d_R
LN_InPec_ESTRO_2d_L	LN_InPec_ESTRO_2d_R	LN_Sclav_ESTRO_2d_L
LN_Sclav_ESTRO_2d_R	LN_Ax_L	LN_Ax_R
LN_Ax_L1_2d_L	LN_Ax_L1_2d_R	LN_Ax_L1_ESTRO_L
LN_Ax_L1_ESTRO_R	LN_Ax_L1_L	LN_Ax_L1_R
LN_Ax_L2_2d_L	LN_Ax_L2_2d_R	LN_Ax_L2_ESTRO_L
LN_Ax_L2_ESTRO_R	LN_Ax_L2_L	LN_Ax_L2_R
LN_Ax_L2_L3_L	LN_Ax_L2_L3_R	LN_Ax_L3_2d_L
LN_Ax_L3_2d_R	LN_Ax_L3_ESTRO_L	LN_Ax_L3_ESTRO_R
LN_Ax_L3_L	LN_Ax_L3_R	LN_Ax_Sclav_2d_L
LN_Ax_Sclav_2d_R	LN_IMN_2d_L	LN_IMN_2d_R
LN_IMN_Expand_2d_L	LN_IMN_Expand_2d_R	LN_IMN_L
LN_IMN_R	LN_IMN_RC_L	LN_IMN_RC_R
LN_InPec_ESTRO_L	LN_InPec_ESTRO_R	LN_Sclav_2d_L
LN_Sclav_2d_R	LN_Sclav_ESTRO_L	LN_Sclav_ESTRO_R
LN_Sclav_L	LN_Sclav_R	LN_Sclav_RC_L
LN_Sclav_RC_R		

Breast — 48

A_Aorta	A_Aorta_Asc	A_Aorta_Dsc
A_LAD	A_Pulmonary	Atrium_L
Atrium_R	BrachialPlex_L	BrachialPlex_R
BrachialPlexs	Breast_Implant_L	Breast_Implant_R

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Breast — 48 (cont.)

Breast_L	Breast_R	Breast_Prone
Breast_Prone_RTOG	Breast_RTOG_L	Breast_RTOG_R
Breast_Wire_L	Breast_Wire_R	Breasts_RTOG
Chestwall_ESTRO_L	Chestwall_ESTRO_R	Chestwall_L
Chestwall_R	Chestwall_RC_L	Chestwall_RC_R
Clavicle_L	Clavicle_R	Esophagus
Glnd_Thyroid	Heart	Heart_Prone
Humerus_L	Humerus_R	Liver
Lung_L	Lung_R	Lungs
Nipple_L	Nipple_R	Nipple_Prone
Pericardium	Stomach	V_Venacava_I
V_Venacava_S	Ventricle_L	Ventricle_R

Mediastinal Nodes — 14

LN_Mediast_1_L	LN_Mediast_1_R	LN_Mediast_2_L
LN_Mediast_2_R	LN_Mediast_3_A	LN_Mediast_3_P
LN_Mediast_4_L	LN_Mediast_4_R	LN_Mediast_5
LN_Mediast_6	LN_Mediast_7	LN_Mediast_8
LN_Mediast_10	LN_Mediast_11	

Thorax — 57

A_Aorta	A_Aorta_Asc	A_Aorta_Dsc
A_Brachiocephls	A_Circumflex_L	A_Coronary_2d_R
A_Coronary_L	A_Coronary_R	A_LAD
A_Pulmonary	A_Subclavian_L	A_Subclavian_R
Atrium_L	Atrium_R	AV_Node
Bone_Sternum	BrachialPlex_L	BrachialPlex_R
BrachialPlexs	Bronchus	Carina
Chestwall_Anat	Chestwall_Anat_L	Chestwall_Anat_R
Chestwall_2cm_L	Chestwall_2cm_R	Chestwall_OAR
Clavicle_L	Clavicle_R	Esophagus
GreatVessels	Heart	Heart_Prone
Heart+A_Pulm	Lung_L	Lung_R
Lungs	Musc_PecMinor_L	Musc_PecMinor_R
Myocardium	Pericardium	Pericardium_Inf
Pericardium_Inf+A_Pulm	Trachea	V_Brachioceph_L
V_Brachioceph_R	V_Pulmonary	V_Venacava_I
V_Venacava_I_UKSABR	V_Venacava_S	V_Venacava_S_UKSABR
Valve_Aortic	Valve_Mitral	Valve_Pulmonic
Valve_Tricuspid	Ventricle_L	Ventricle_R

Abdomen — 25

A_Aorta	A_Aorta_Dsc	A_Celiac
A_Mesenteric_S	BileDuct_Common	Bowel
Bowel_Bag	Bowel_Large	Bowel_Small
Duodenum	Esophagus	Gallbladder
Kidney_L	Kidney_R	Kidney_Outer_L
Kidney_Outer_R	Kidneys	Kidneys_Outer
Liver	LN_Paraaortic	Pancreas
Spleen	Stomach	V_Venacava_I
V_Venacava_I_UKSABR		

Pelvis (Male) — 60

A_Aorta	A_Aorta_Dsc	Barrigel
Bladder	Bone_Ilium	Bone_Ilium_L
Bone_Ilium_R	Bone_Ischium_L	Bone_Ischium_R

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Pelvis (Male) — 60 (cont.)

Bone_Pelvic	Bone_Sacrum	Bowel
Bowel_Bag	Bowel_Large	Bowel_Small
Canal_Anal	Colon_Sigmoid	Femur_Head_L
Femur_Head_R	Femur_L	Femur_R
Femur_RTOG_L	Femur_RTOG_R	Femurs_RTOG
Genitals_M	Iliac_Int_L	Iliac_Int_R
Iliac_L	Iliac_R	LN_Inguinofem_L
LN_Inguinofem_R	LN_Mesorectum	LN_Paraaortic
LN_Pelvics	LN_Pelvics_NRG	LN_Presacral
LN_Rectum_EIN_L	LN_Rectum_EIN_R	Marrow_Ilium_L
Marrow_Ilium_R	Marrow_Pelvis	Musc_Iliopsoas_L
Musc_Iliopsoas_R	PenileBulb	Prostate
Prostate-Spacergel_VUE	Prostate+SeminalVes	Prostate+SV-Spacergel_VUE
ProstateBed	ProstateFiducials	Prostate-Gel_Fiducials
Bone_PubicSymphys	Rectum	Rectum-Spacergel_VUE
SpaceOARVue	SacralPlex_L	SacralPlex_R
SeminalVes	V_Venacava_I	Vessels_Pelvis

Pelvis (Female) — 50

A_Aorta	A_Aorta_Dsc	Bladder_F
Bone_Ilium	Bone_Ilium_L	Bone_Ilium_R
Bone_Ischium_L	Bone_Ischium_R	Bone_Pelvic
Bone_Sacrum	Bowel_Bag_F	Bowel_F
Bowel_Large_F	Bowel_Small_F	Canal_Anal_F
Colon_Sigmoid	Femur_Head_L	Femur_Head_R
Femur_L	Femur_R	Femur_RTOG_L
Femur_RTOG_R	Femurs_RTOG	Genitals_F
Iliac_Int_L	Iliac_Int_R	Iliac_L
Iliac_R	LN_Pelvics_F	LN_Presacral
LN_Inguinofem_L	LN_Inguinofem_R	LN_Paraaortic
LN_Mesorectum	LN_Rectum_EIN_L	LN_Rectum_EIN_R
Marrow_Ilium_L	Marrow_Ilium_R	Marrow_Pelvis
Musc_Iliopsoas_L	Musc_Iliopsoas_R	Parametrium
Bone_PubicSymphys	Rectum_F	SacralPlex_L
SacralPlex_R	UteroCervix	Vagina
V_Venacava_I	Vessels_Pelvis	

HDR Female Pelvis — 10

HDR_Bladder	HDR_Bowel	HDR_Canal_Anal
HDR_Cylinder	HDR_Rectum	HDR_Ring
HDR_Colon_Sigmoid	HDR_Urethra	HDR_UteroCervix
Foley_Balloon		

Brain MR — 42

Amygdala_L	Amygdala_R	Brain
Brainstem	CaudateNucleus_L	CaudateNucleus_R
Cerebellum	Cornea_L	Cornea_R
CorpusCallosum	Eye_L	Eye_R
Falx	Fornix_L	Fornix_R
Glnd_Lacrimal_L	Glnd_Lacrimal_R	Hippocampus_L
Hippocampus_R	Hypo_True	Hypothalamus
Lens_L	Lens_R	Medulla
Midbrain	OpticChiasm	OpticNrv_L
OpticNrv_R	Optics	OpticTract_L
OpticTract_R	Periventricular_Space	Pineal
Pituitary	Pons	Retina_L

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Brain MR — 42 (cont.)

Retina_R	Sinuses	SpinalCord_Cerv
Tentorium	Thalamus	Ventricle_Brain

Pelvis (Male) MR — 19

A_Pud_Int_L	A_Pud_Int_R	Bladder
Bladder_Trigone	Colon_Sigmoid	External_Pelvis
Femur_L	Femur_R	Glnd_Prostate
NVB_L	NVB_R	PenileBulb
Prostate	PubicSymphys	Rectal_Spacer
Rectum	Bone_Sacrum	SeminalVes
Urethra		

Pelvis (Male) MR TRUFI — 2

Bladder_TRUFI	PenileBulb_TRUFI

HDR Female Pelvis MR — 7

HDR_Bladder	HDR_Bowel	HDR_Canal_Anal
HDR_Colon_Sigmoid	HDR_Rec-Canal	HDR_Rectum
HDR_Urethra		

Abdomen MR TRUFI — 11

Bowel	Duodenum	Esophagus
Gallbladder	Kidney_L	Kidney_R
Liver	Pancreas	Spleen
Stomach	ThecalSac	

Pelvis (Male) CBCT — 7

Bladder_CBCT	Femur_L_CBCT	Femur_R_CBCT
LN_Pelvics_CBCT	Prostate_CBCT	Rectum_CBCT
SeminalVes_CBCT		

MultiSite — 58

Body	CaudaEquina	External
Rib	Rib_L	Rib_R
Rib01_L	Rib01_R	Rib02_L
Rib02_R	Rib03_L	Rib03_R
Rib04_L	Rib04_R	Rib05_L
Rib05_R	Rib06_L	Rib06_R
Rib07_L	Rib07_R	Rib08_L
Rib08_R	Rib09_L	Rib09_R
Rib10_L	Rib10_R	Rib11_L
Rib11_R	Rib12_L	Rib12_R
Skin	SpinalCanal	SpinalCord
VB	VB_C1	VB_C2
VB_C3	VB_C4	VB_C5
VB_C6	VB_C7	VB_L1
VB_L2	VB_L3	VB_L4
VB_L5	VB_T01	VB_T02
VB_T03	VB_T04	VB_T05
VB_T06	VB_T07	VB_T08
VB_T09	VB_T10	VB_T11
VB_T12		

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Evidence & Limitations

Clinical Evidence: N/A

Limitations: N/A

Evidence by Source (8)

Geometric and dosimetric evaluation for breast and regional nodal auto-segmentation structures

Tsui T et al. · J Appl Clin Med Phys · 2024 · DOI 10.1002/acm2.14461

E1 L2

Single-centre retrospective geometric and dosimetric validation of AutoContour for breast and regional nodes.

https://doi.org/10.1002/acm2.14461

Artificial Intelligence in Radiation Treatment Planning: A Survey-Based Observational Study From A Cancer Centre in Nigeria

Alabi AO et al. · Technol Cancer Res Treat · 2026 · DOI 10.1177/15330338251403601

E1 L3 Sets product level

Survey-based observational study of AutoContour users reporting perceived workflow impact.

https://doi.org/10.1177/15330338251403601

Evaluation of a prostate SBRT planning workflow using auto segmentation and

Jones T et al. · Journal of Applied Clinical Medical Physics · 2025 · DOI 10.1002/acm2.70295

E1 L2

Single-centre prostate SBRT workflow with AutoContour and RapidPlan; dosimetric endpoints.

https://doi.org/10.1002/acm2.70295

Plan Quality Analysis of Automated Treatment Planning Workflow With Commercial Auto-Segmentation Tools and Clinical Knowledge-Based Planning Models for Prostate Cancer

Adams J et al. · Cureus · 2023 · DOI 10.7759/cureus.41260

E1 L2

Single-centre plan-quality comparison using AutoContour and AI-Rad Companion structure sets.

https://doi.org/10.7759/cureus.41260

Auto-contouring comparison across 5 facilities including Radformation

Goddard L et al. · Radiother Oncol · 2024 · DOI 10.1186/s13014-024-02451-4

E2 L2 Sets product level

Multi-centre evaluation across five facilities including review and edit effort.

https://doi.org/10.1186/s13014-024-02451-4

A clinical evaluation of the performance of five commercial artificial intelligence contouring systems for radiotherapy

Doolan PJ et al. · Front. Oncol. · 2023 · DOI 10.3389/fonc.2023.1213068

E2 L3

Independent benchmark of several commercial systems on external clinical data; geometric endpoints.

https://doi.org/10.3389/fonc.2023.1213068

For information only. Not a clinical validation or endorsement; verify with the manufacturer and local commissioning.

Evaluation and failure analysis of four commercial deep learning based autose
abdominal organs at risk

Fan M et al. · J Applied Clin Med Phys · 2025 · DOI 10.1002/acm2.70010

E2 L2

Vendor-independent evaluation and failure analysis of four commercial systems on abdominal OARs; Radformation explicitly included.
https://doi.org/10.1002/acm2.70010

Investigation on performance of multiple AI-based auto-contouring systems in organs at risks (OARs) delineation

Kim YW et al. · Phys Eng Sci Med · 2024 · DOI 10.1007/s13246-024-01434-9

E2 L2

Independent comparative benchmark of seven commercial systems including Radformation; geometric endpoints.
https://doi.org/10.1007/s13246-024-01434-9

Regulatory Information

CE: cleared FDA: 510k_cleared

CE Certification:	cleared
CE Details:	cleared (Class IIa)
FDA Status:	510k_cleared
FDA Details:	510k_cleared (K262452)
Intended Use Statement:	AutoContour is intended to assist radiation treatment planners in contouring and reviewing structures within medical images in preparation for radiation therapy treatment planning. (Source: FDA 510(k) K262452 Summary, accessed 2026-09-22)

Market Information

Market Presence:	2020
On Market Since:	2020
Distribution Channels:	Direct sales; Integrated cloud contouring service; Includes Limbus AI (acquisition announced April 30, 2024)

Contact Information

Website:	https://www.radformation.com/autocontour/autocontour
Company URL:	https://www.radformation.com/
Product URL:	https://www.radformation.com/autocontour/autocontour
Logo URL:	/logos/radformation.svg
Logo Source:	Logo source: /logos/radformation.svg

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Support Email:	N/A

Quality Assurance

Last Revised:	9/26/2026
Company Revision Date:	N/A
Source:	FDA 510(k) database (K260509, cleared 2026-03-19; K262452, cleared 2026-08-14), FDA summaries, Radformation product page and EU AI/MDR compliance page, BusinessWire MDR CE Mark release, peer-reviewed validation studies (Doolan 2023; Goddard 2024; Kim 2024; Lin/Fan 2025) — all four explicitly include Radformation per their Methods sections (re-verified 2026-06-15). v2.8 features and 520+ model count from Radformation blog 'AutoContour v2.8' (2026-09-25, https://blog.radformation.com/autocontour-v2.8-broader-coverage-smarter-automation-and-more-plan-ready-workflows, retrieved 2026-09-26); the announcement does not state which clearance covers v2.8. Per-structure list re-synced to v2.8 from the Radformation AutoContour product page (https://www.radformation.com/autocontour/autocontour, retrieved 2026-09-26): 17 model groups whose listed counts (607 entries in total, with left/right counted separately) match the extracted names exactly; the same structure appearing in several groups is listed under each, giving 523 distinct models (435 CT, 81 MR, 7 CBCT), consistent with the vendor's '520+'. The previous v2.7 list is kept under structureHistory. No v2.8 model is labelled as a target volume: neither the product page nor the K262452 summary names any CTV/GTV/PTV, and the summary describes the models as trained on 'anatomical organs'. Guideline-named breast/chest-wall volumes (e.g. Breast_RTOG_L/R, Chestwall_ESTRO_L/R, Chestwall_RC_L/R) are therefore kept as organs; the v2.7 list had normalised two of them to 'CTV Breast L/R'.
GitHub URL:	https://github.com/DLinRT-eu/dlinrteu-website/tree/main/src/data/products/auto-contouring/radformation.ts

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